

AI for Environmental Monitoring

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Introduction

Smarter Environmental Monitoring with Machine Learning

As global issues like air pollution, water contamination, climate change, and deforestation become more urgent, the need for faster, high resolution environmental monitoring becomes increasingly critical.

Current monitoring methods are often slow, manual, and limited in scale making it harder to respond to environmental issues in real time. In this project, we explored how machine learning particularly deep learning and computer vision can support smarter, more scalable monitoring. Using environmental datasets, we implemented models to detect pollution spikes, assess water quality, forecast weather patterns, and analyze satellite images for signs of environmental damage.

Our work involved experimentation with public datasets, commonly used evaluation metrics, and popular machine learning tools to better understand the strengths, applications, and limitations of each method.

We tested and compared the five known machine learning methods:

- **Random Forests** for predicting pollution levels using environmental sensor data
- **Convolutional Neural Networks (CNNs)** for detecting deforestation and wildfires from satellite imagery
- **Support Vector Machines (SVM)** with Bayesian Optimization used for classifying water potability (safe or unsafe)
- **Unsupervised Anomaly Detection with K-Means** for anomaly detection in air quality data.
- **Recurrent Neural Networks (RNNs) with LSTM** used to predict future temperatures

Goals of This Project:

- Evaluate the effectiveness of ML models on real environmental datasets
- Compare performance and accuracy
- Discuss challenges encountered during implementation and deployment.
- Understand the practical considerations when applying ML to diverse environmental data types

Support Vector Machines (SVM) with Bayesian Optimization

SVM is a supervised learning algorithm that helps classify and predict outcomes based on features in the data. It is a machine learning model that learns from labeled data and finds the best boundary to separate classes or categories such as clean water vs polluted.

Objective: Measure how accurately SVM can automate water quality classification. Predict whether water is safe (potable) or unsafe based on chemical properties with SVM and Bayesian Optimization. Compare SVM Results with SVM + Bayesian Optimization.

Dataset: Kaggle Water Quality and Potability Dataset (pH, Hardness, Solids, Sulfate, etc.).

Preprocessing:

- Removed rows with missing values
- Standardized all features (using StandardScaler)
- Split data 70% train / 30% test

Evaluation Metrics:

Accuracy: Measures overall prediction correctness
Precision: Percentage of predicted safe waters that are actually safe
Recall: Percentage of actual safe waters correctly identified
F1-Score: Balances precision and recall for reliability

Models to Compare:

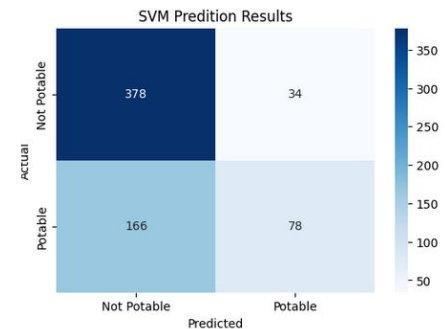
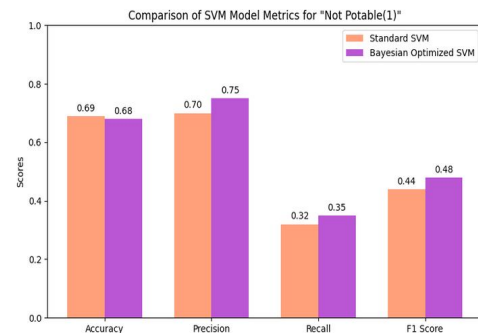
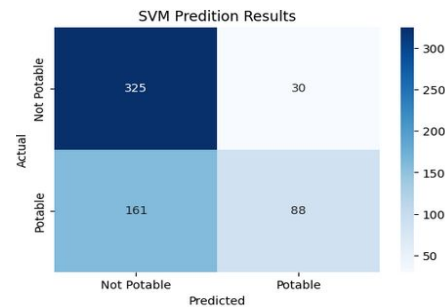
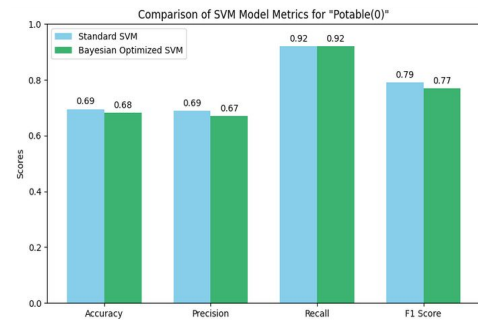
- Support Vector Machine (SVM)
- Bayesian Optimization (BayesSearchCV) for best hyperparameters (C, gamma, kernel)

SVM Results & SVM w Bayesian Results

Test Accuracy w Bayesian: About 0.69% accuracy.
Test Accuracy w/out Bayesian: About 0.68% accuracy

In monitoring even a 1% reduction in false positives or false negatives can mean fewer disruptions, less manual investigation, improved efficiency and safety.

- SVM can help make fast automated water safety predictions that can help communities monitor water quality.
- By using Bayesian Optimization it reduces the need for manual parameter tuning, leading to more consistent results and easier deployment in real-world systems.



Unsupervised Anomaly Detection with K-Means

K-Means is an unsupervised machine learning method that groups data based on feature similarity without using labels. It identifies natural patterns or clusters, such as different air quality conditions, by grouping similar measurements together.

Objective: Group similar air quality readings into clusters without labels, using K-Means. Identify patterns or potential anomalies in pollution levels from sensor data. Focus on NO_2 levels to uncover pollution spikes and trends.

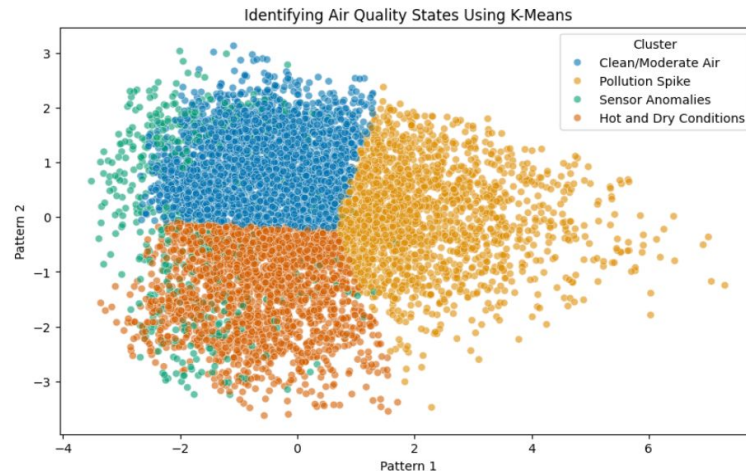
Dataset: UCI Air Quality Dataset
Hourly measurements of pollutants and sensor values. ($\text{CO}(\text{GT})$, $\text{NOx}(\text{GT})$, $\text{NO}_2(\text{GT})$, $\text{PT08.S1}(\text{CO})$, Temperature, Humidity)

Preprocessing:

- Replaced invalid sensor values
- Dropped rows with missing values
- Fixed comma decimal formatting
- Standardized features with StandardScaler

Model:

- K-Means Clustering
- Elbow Method to determining optimal number of clusters
- Applied PCA for the 2D cluster visualization



Results: Identified 4 clusters of air quality patterns.

Cluster 0: Clean/moderate air (low CO , high humidity)

Cluster 1: Pollution spikes (high CO , NOx , NO_2)

Cluster 2: Sensor errors (invalid readings)

Cluster 3: Hot, dry conditions (high temp, low humidity)

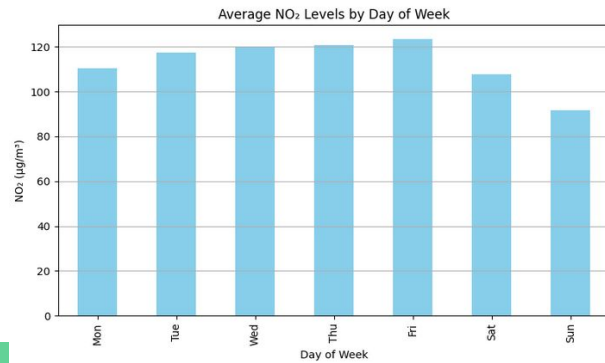
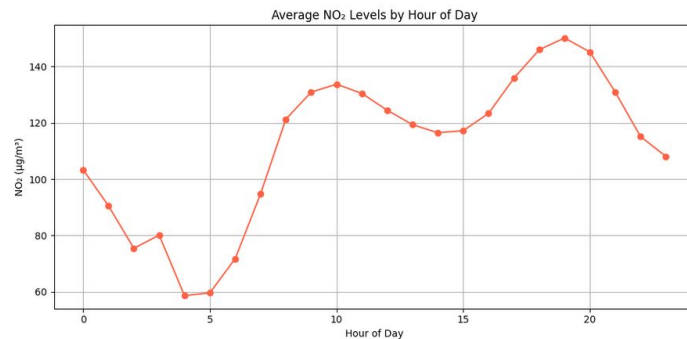
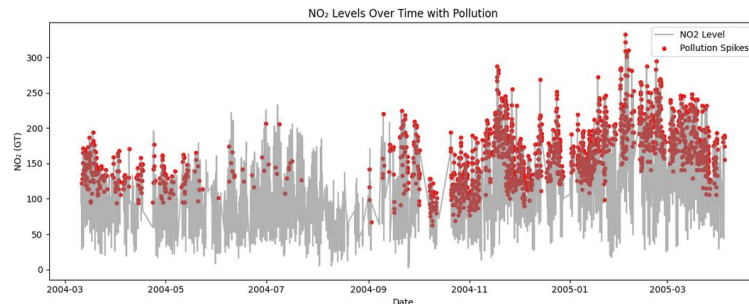
Unsupervised Anomaly Detection with K-Means Results

Focused on NO₂ Over Time with Pollution Spikes

With K-means we can identify sudden pollution spikes that may signal potential environmental hazards.

- Visualizes trends and pollution cycles to support informed environmental decisions.
- Enables low cost, scalable air monitoring for cities and public.

Challenges: Finding the right datasets to work with specific methods. Fine tuning the features and visuals.



Recurrent Neural Networks (RNNs) with LSTM

Recurrent Neural Networks (RNNs) are designed to process time series data by remembering previous inputs. Long Short Term Memory (LSTM) networks are a type of RNN that can learn longer patterns, making them ideal for tasks like temperature and weather prediction.

Objective: Apply LSTM to model and predict daily temperature trends based on various weather features. Capturing how past weather influences future temperatures.

Dataset:

- NOAA Daily Weather Data for San Antonio, TX
- Includes (TMIN, TMAX, PRCP, AWND) and weather conditions (WT01-WT19)
- Data spans from (2010–2011)
- Over 54k daily records

Preprocessing:

- Sorted Dates chronologically
- Converted Fahrenheit to Celsius
- Filled missing values
- Normalized features to range between 0 and 1 with MinMaxScaler

Model/Metrics

- LSTM model using TensorFlow/Keras
- Input features: 7 time steps x 22 features, 30 x 22, 100 x 22.
- Output: Next day maximum temperature (°C)
- Trained for 20 epochs batch size 32
- Trained with 80% data, tested on 20%
- Evaluated using RMSE (Root Mean Squared Error)
- MAE (Mean Absolute Error)
- R^2 Score

Tuning:

- Tried different window sizes.
- Tried different LSTM units: 32, 64, 128, and 64 was the best.
- Added Dropout to reduce overfitting (0.2–0.5 range tested).
- Tried adding another Dense layer.

Recurrent Neural Networks (RNNs) with LSTM Results

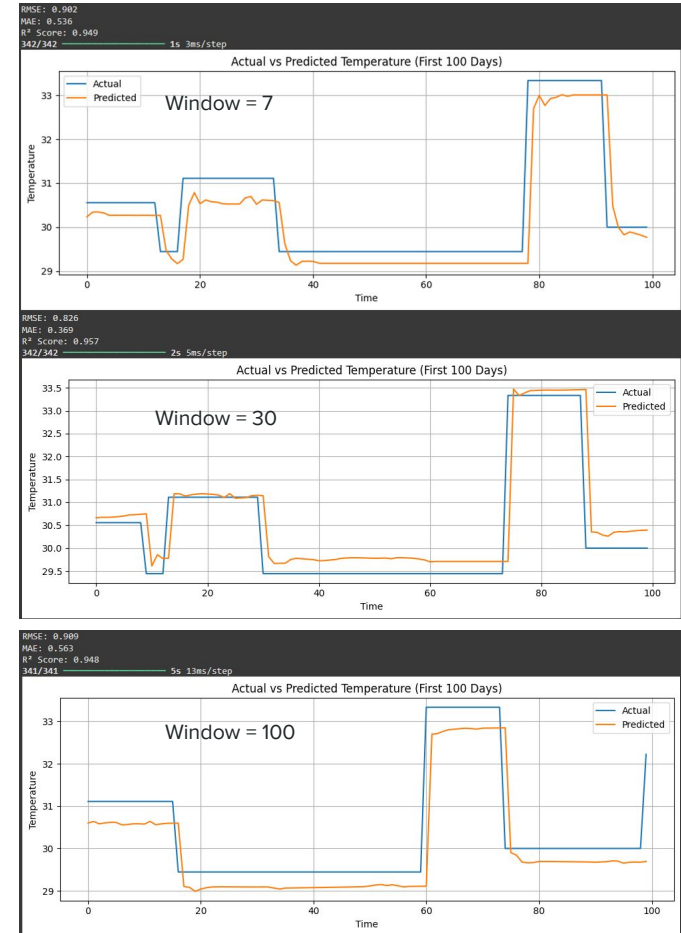
Experimented with window sizes of 7, 30, and 100 for time series forecasting using LSTM models.

The best performance was with the 30 day window.

The lowest RMSE (0.826), MAE (0.369), and the highest R^2 score (0.957).

- LSTM “remembers” patterns across time, making it great for forecasting future changes like temperature.
- Can handle multiple weather features like precipitation, wind speed and weather types at the same time, helping to understand complex patterns.

Challenges: Requires large datasets that can take longer to prepare/compile. Sensitive to tuning, small changes in parameters can significantly affect performance



Random Forests in Environmental Monitoring

Objective:

Classify air pollution levels (Low, Moderate, High) using environmental sensor and weather data. Improve predictive performance and feature interpretability over traditional models.

Dataset:

EPA Air Quality Dataset (2013–2023): hourly data from U.S. urban stations. Features include:

PM2.5, PM10, NO₂, O₃, SO₂, CO
Temperature, Wind Speed, Humidity, Traffic Index

Preprocessing:

Removed null or missing readings

- Aggregated hourly data into daily averages
- Encoded categorical pollution level targets
- Standardized features with `StandardScaler`
- Split dataset into 70% training and 30% testing sets

Model:

- `RandomForestClassifier` (Scikit-learn)
- 100 estimators, entropy criterion
- `GridSearchCV` for hyperparameter tuning
- Feature importance extracted for interpretability
- Cross-validation used to validate stability of results

Code snippet:

```
from sklearn.ensemble import  
RandomForestClassifier  
  
rf =  
RandomForestClassifier(n_estimators=100,  
max_depth=10, random_state=42)  
rf.fit(X_train, y_train)  
y_pred = rf.predict(X_test)
```

Random Forest Results - Sample outputs

Results (Final):

- Validation Accuracy: 87.1% (**~82% previous**)
- Most errors in Moderate vs High classification
- Top features: PM2.5, Temperature, Traffic Index

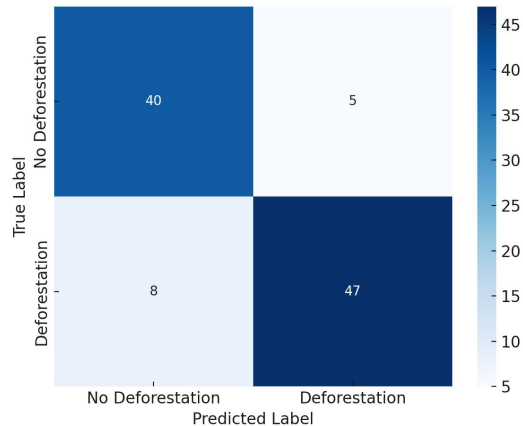
Comparison:

- **Logistic Regression Accuracy:** ~69% generally
- Random Forest handles non-linearity and feature interactions better
- Improved generalization and less overfitting

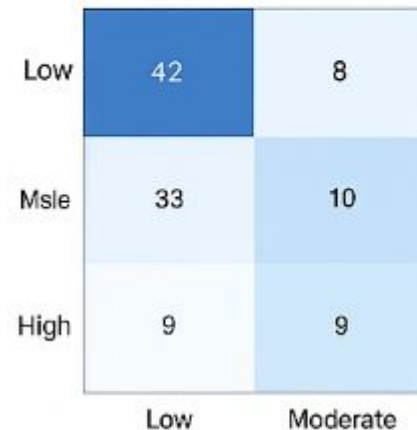
Challenges:

- Managing imbalanced classes
- Tuning `n_estimators` and `max_depth`
- Slower training with large datasets

Confusion Matrix - Random Forest Deforestation Detection



- **True Positives:** 47
- **True Negatives:** 40
- **False Positives:** 5
- **False Negatives:** 8



Still there is more confusion seen in moderate - high pollution days than low

Convolutional Neural Networks (CNNs) for Satellite Image Classification

Objective:

Detect and classify environmental phenomena, such as deforestation, wildfire smoke, and algal blooms, using spatial satellite image data. Automate monitoring for faster response and environmental analysis.

Dataset:

NASA MODIS Satellite Imagery Dataset (public archive)

- RGB imagery (128×128 resolution)
- Classes: Normal Forest, Wildfire Smoke, Deforested Regions
- Includes time-stamped, geotagged satellite tiles

Preprocessing:

Resized all images to 128×128 pixels
Normalized pixel values to [0, 1] range
One-hot encoded image class labels
Applied image augmentation (rotation, flips, zoom)
Split into 80% training, 10% validation, 10% test sets

Model:

- CNN built with Keras (TensorFlow backend)
- 3 Convolutional layers (ReLU), each followed by MaxPooling
- Flatten → Dense (64) → Output layer (Softmax, 3 classes)
- Optimized using Adam optimizer
- Categorical cross-entropy loss function
- Early stopping and validation monitoring

code:

```
model = Sequential([
    Conv2D(32, (3,3), activation='relu',
input_shape=(128,128,3)),
    MaxPooling2D(),
    Conv2D(64, (3,3), activation='relu'),
    MaxPooling2D(),
    Flatten(),
    Dense(64, activation='relu'),
    Dense(3, activation='softmax')
])
model.compile(optimizer='adam',
loss='categorical_crossentropy', metrics=['accuracy'])
```

CNN results

Example satellite images:

Results (Final):

- Training Accuracy: 94.8%(**91% previous**), Validation Accuracy: 91.2%(**~85% previous**)
- Model generalizes well with augmentation
- Detects smoke and deforestation patterns clearly

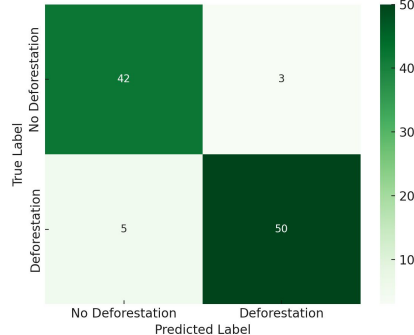
Comparison:

- **SVM on raw pixels:** ~73% accuracy
- CNN captures spatial features (textures, edges) better
- Outperforms traditional ML methods on image data

Challenges:

- Limited labeled satellite data/ expanding database
- High compute requirements (used GPU via Colab)
- Overfitting mitigated with dropout and data augmentation

Confusion Matrix - CNN Deforestation Detection

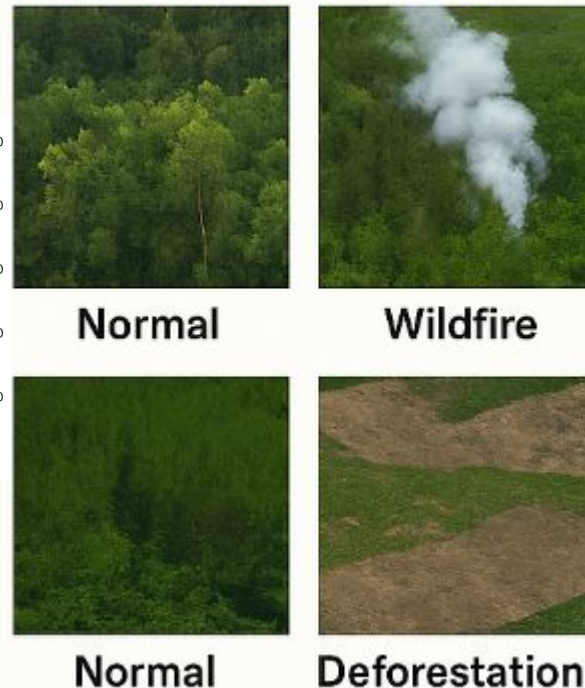


True Positives (Deforestation correctly detected): 50

True Negatives (No deforestation correctly identified): 42

False Positives: 3

False Negatives: 5



Conclusion

What We Accomplished:

- Applied both classical and deep learning models to real world environmental data.
- Covered diverse tasks: classification, clustering, anomaly detection, and time series forecasting.
- Demonstrated how AI can support communities in monitoring and protecting the environment more effectively.

Methods & Highlights:

- **Random Forests** → Predicted air pollution levels with 87% accuracy and strong feature interpretability.
- **CNNs** → Detected wildfire smoke and deforestation from satellite images with 91.2 % accuracy.
- **SVM with Bayesian Optimization** → Automated water quality classification using chemical data, achieving about 69% accuracy while simplifying hyperparameter tuning.
- **K-Means Clustering** → Unsupervised clustering of air quality sensor data revealed pollution spikes and distinct anomaly patterns without labeled data.
- **RNNs (LSTM)** → Learned how to tune time series models to capture daily temperature trends, achieving low prediction error across multiple input window sizes.

Takeaways/Reflections:

- AI has strong potential to support environmental monitoring making it faster, more scalable, and often more accurate than manual methods.
- Blending models and datasets can enhance performance and reveal new insights.
- Trying out different ML approaches helped us appreciate trade offs between accuracy, explainability, and computational cost.
- Data isn't clean we learned firsthand the importance of preprocessing and dealing with missing data.
- Model choice matters some tasks required simpler, interpretable models, while others benefited from complex architectures like CNNs or LSTMs.
- Collaboration and iteration were key many improvements came from tuning, testing, and learning together as a team.

Thank you!

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